5. Hospital Patient Data Analysis

Scenario

A hospital maintains records of patients including age, gender, diagnosed disease, assigned doctor, and treatment cost.

Hospital management wants to analyze patient data to identify disease patterns and doctor workload.

They want to understand:

- Which diseases are most common?
- Which doctors treat the most patients?
- What is the average treatment cost?

Students must analyze healthcare data to provide insights.

Sample CSV

patient_id,age,gender,disease,doctor,treatment_cost
1,45,Male,Diabetes,Dr Shah,5000
2,30,Female,Fever,Dr Patil,2000
3,55,Male,Heart Disease,Dr Shah,20000
4,22,Female,Flu,Dr Kulkarni,1500
5,40,Male,Diabetes,Dr Shah,4500

SQL Tasks

Students should:

- Display patient records.

```
mysql> SELECT * FROM patients;
+------------+------+--------+---------------+-------------+----------------+
| patient_id | age  | gender | disease       | doctor      | treatment_cost |
+------------+------+--------+---------------+-------------+----------------+
|          1 |   45 | Male   | Diabetes      | Dr Shah     |           5000 |
|          2 |   30 | Female | Fever         | Dr Patil    |           2000 |
|          3 |   55 | Male   | Heart Disease | Dr Shah     |          20000 |
|          4 |   22 | Female | Flu           | Dr Kulkarni |           1500 |
|          5 |   40 | Male   | Diabetes      | Dr Shah     |           4500 |
+------------+------+--------+---------------+-------------+----------------+
5 rows in set (0.10 sec)
```

- Count patients by disease.

```
mysql> SELECT disease, COUNT(*) AS patient_count FROM patients GROUP BY disease;
+---------------+---------------+
| disease       | patient_count |
+---------------+---------------+
| Diabetes      |             2 |
| Fever         |             1 |
| Heart Disease |             1 |
| Flu           |             1 |
+---------------+---------------+
4 rows in set (0.01 sec)
```

- Calculate average treatment cost.

```
mysql> SELECT AVG(treatment_cost) AS avg_cost FROM patients;
+-----------+
| avg_cost  |
+-----------+
| 6600.0000 |
+-----------+
1 row in set (0.01 sec)
```

- Identify doctor treating most patients.

```
mysql> SELECT doctor, COUNT(*) AS total_patients FROM patients GROUP BY doctor ORDER BY total_patients DESC LIMIT 1;
+---------+----------------+
| doctor  | total_patients |
+---------+----------------+
| Dr Shah |              3 |
+---------+----------------+
1 row in set (0.01 sec)
```

Excel Tasks

Students should create pivot table:

Rows → Disease
Values → Patient count

	disease	Patient Count
16	disease	Patient Count
17	Diabetes	6
18	Fever	2
19	Flu	4
20	Heart Disease	3
21	**Grand Total**	**15**

Python Tasks

Students must:

- Load dataset.
- Calculate disease frequency.
- Calculate treatment cost statistics.
- Group patients by doctor.

Python code:-

```
import pandas as pd

# Load dataset

df = pd.read_csv("patients.csv")

print("===== FIRST 5 ROWS =====")

print(df.head())

print("\n===== DATASET INFO =====")

print(df.info())
```

```
print("\n===== SUMMARY STATISTICS =====")
print(df.describe())
# Disease frequency
print("\n===== DISEASE COUNT =====")
print(df["disease"].value_counts())
# Doctor workload
print("\n===== DOCTOR WORKLOAD =====")
print(df["doctor"].value_counts())
# Average treatment cost
avg_cost = df["treatment_cost"].mean()
print("\nAverage Treatment Cost:", avg_cost)
# Highest cost case
print("\n===== HIGHEST COST PATIENT =====")
print(df.loc[df["treatment_cost"].idxmax()])
# Age group analysis (extra insight)
df["age_group"] = pd.cut(
    df["age"],
    bins=[0, 30, 50, 100],
    labels=["Young", "Adult", "Senior"]
)
print("\n===== AGE GROUP DISTRIBUTION =====")
print(df["age_group"].value_counts())
```

```
# Save processed data (optional for Excel use)

df.to_csv("patients_processed.csv", index=False)

print("\nProcessed file saved as 'patients_processed.csv'")
```

```
===== FIRST 5 ROWS =====
   patient_id  age  gender        disease       doctor  treatment_cost
0           1   45    Male       Diabetes      Dr Shah            5000
1           2   30  Female          Fever     Dr Patil            2000
2           3   55    Male  Heart Disease      Dr Shah           20000
3           4   22  Female            Flu  Dr Kulkarni            1500
4           5   40    Male       Diabetes      Dr Shah            4500

===== DATASET INFO =====
<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   patient_id      5 non-null      int64
 1   age             5 non-null      int64
 2   gender          5 non-null      str
 3   disease         5 non-null      str
 4   doctor          5 non-null      str
 5   treatment_cost  5 non-null      int64
dtypes: int64(3), str(3)
memory usage: 372.0 bytes
None
```

```
===== SUMMARY STATISTICS =====
       patient_id        age  treatment_cost
count    5.000000   5.000000        5.000000
mean     3.000000  38.400000     6600.000000
std      1.581139  12.856905     7643.624795
min      1.000000  22.000000     1500.000000
25%      2.000000  30.000000     2000.000000
50%      3.000000  40.000000     4500.000000
75%      4.000000  45.000000     5000.000000
max      5.000000  55.000000    20000.000000

===== DISEASE COUNT =====
disease
Diabetes         2
Fever            1
Heart Disease    1
Flu              1
Name: count, dtype: int64

===== DOCTOR WORKLOAD =====
doctor
Dr Shah        3
Dr Patil       1
Dr Kulkarni    1
Name: count, dtype: int64

Average Treatment Cost: 6600.0
```

```
===== HIGHEST COST PATIENT =====
patient_id                    3
age                          55
gender                     Male
disease           Heart Disease
doctor                  Dr Shah
treatment_cost            20000
Name: 2, dtype: object

===== AGE GROUP DISTRIBUTION =====
age_group
Young     2
Adult     2
Senior    1
Name: count, dtype: int64

Processed file saved as 'patients_processed.csv'
```

Output of python code

	A	B	C	D	E	F	G	H
1	patient_id	age	gender	disease	doctor	treatmer	age_group	
2	1	45	Male	Diabetes	Dr Shah	5000	Adult	
3	2	30	Female	Fever	Dr Patil	2000	Young	
4	3	55	Male	Heart Disease	Dr Shah	20000	Senior	
5	4	22	Female	Flu	Dr Kulkarni	1500	Young	
6	5	40	Male	Diabetes	Dr Shah	4500	Adult	
7					Average Treatment Cost	6600		
8								

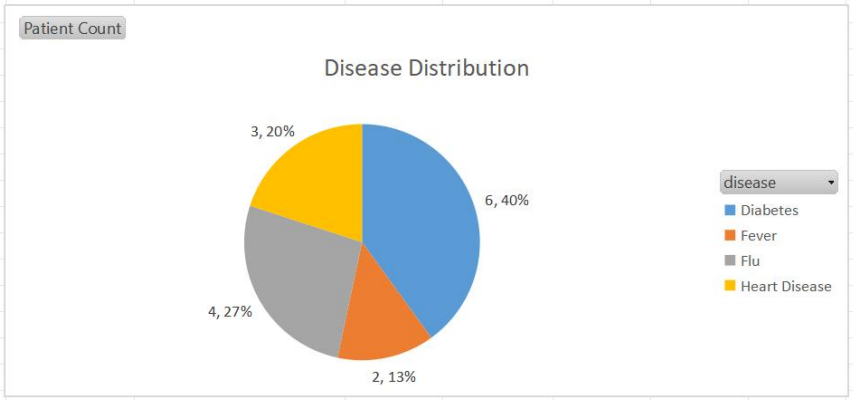
Patients_processed.csv file

Simple Visualization Tasks

Students should create:

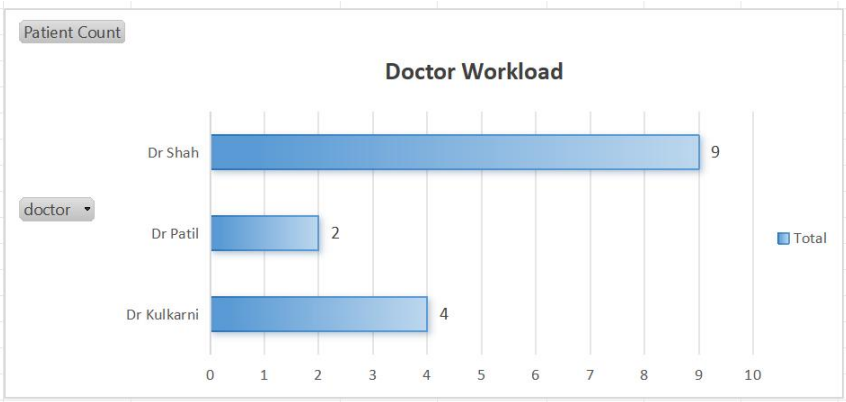
- Pie chart → Disease distribution

disease	Patient Count
Diabetes	6
Fever	2
Flu	4
Heart Disease	3
Grand Total	**15**



- Bar chart → Doctor workload

doctor	Patient Count
Dr Kulkarni	4
Dr Patil	2
Dr Shah	9
Grand Total	**15**



Conclusion

The analysis helped identify disease patterns, doctor workload, and treatment costs, enabling better decision-making and efficient hospital resource management.